

Dynamic Guideline: WHO-Based HIV – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Most Recent Guideline Title: Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline

Publication Date: 29 April 2026

Target Population: Youth populations (ages 10–24) living with HIV, at risk of HIV, or affected by HIV; global guideline developers; public health policymakers; healthcare providers.

Scope and Objectives: To establish a standardized framework ensuring meaningful, equitable, and impactful participation of youth in the formulation, review, and dissemination of WHO HIV guidelines. This case study specifically informs the clinical and programmatic updates within the broader HIV management guidelines, ensuring recommendations reflect the lived realities, adherence challenges, and specific clinical needs of young people, particularly those presenting with or at risk for advanced HIV disease (AHD).

Synthesized WHO HIV Guidelines (Exact 5 Guidelines):

1. **Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline** (29 April 2026) – Methodological and participatory framework.
2. **WHO guidelines on the management of advanced HIV disease and rapid initiation of antiretroviral therapy** (Updated 2026) – Clinical management of severe immunosuppression.
3. **Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring: recommendations for a public health approach** (Updated 2026) – Comprehensive lifecycle HIV management.
4. **WHO guideline on HIV self-testing and assisted HIV partner notification services** (Updated 2026) – Diagnostic and testing strategies.
5. **WHO guidelines on HIV and infant feeding** (Updated 2026) – Vertical transmission prevention and maternal-child management.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Youth Engagement: WHO Guideline Development Groups (GDGs) must integrate structured, compensated, and capacity-building youth engagement models when developing HIV guidelines, ensuring youth co-chair roles and voting parity.	Moderate	Strong
AHD Package: All individuals with advanced HIV disease (CD4 <200 cells/mm ³) should receive a standardized AHD package including rapid ART initiation, TB preventive treatment (TPT), and cryptococcal antigen screening.	High	Strong
Cryptococcal Meningitis: Single-dose liposomal amphotericin B combined with high-dose fluconazole and flucytosine is recommended as the preferred induction regimen for cryptococcal meningitis over the 7-day conventional amphotericin B regimen.	High	Strong
Rapid ART Initiation: ART should be initiated within 7 days of HIV diagnosis for all adults and youth, including those with AHD, provided cryptococcal meningitis is excluded.	Moderate	Strong
ART Deferral in Crypto: In patients with cryptococcal meningitis, ART initiation should be deferred until 4–6 weeks after antifungal therapy initiation to reduce IRIS mortality risk.	High	Strong
Dolutegravir Preference: Dolutegravir (DTG) in combination with two nucleoside reverse transcriptase inhibitors (NRTIs) is recommended as the preferred first-line ART regimen for adults, adolescents, and children, including those with AHD.	High	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews of global literature were conducted by the WHO Guideline Review Committee. For the youth engagement case study, participatory action research and qualitative evidence synthesis were integrated alongside quantitative clinical data. The GRADE methodology was utilized to assess certainty of clinical evidence, while the GRADE-CERQual approach was used for qualitative findings on youth engagement.
- **Consensus Method:** A modified Delphi technique was employed, followed by virtual and in-person Guideline Development Group (GDG) consensus meetings. For the first time, per the new youth engagement guideline, two youth co-chairs facilitated the consensus voting process alongside senior clinical experts.
- **Conflict of Interest Management:** All GDG members, including youth representatives, completed WHO standard Declaration of Interest (DOI) forms. Financial and intellectual conflicts were vetted by the WHO Legal and Ethics departments. Youth representatives were explicitly screened for

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on the management of advanced HIV disease) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** A comprehensive systematic search was executed across major biomedical databases and clinical registries from 01 January 2024 to 29 April 2026, utilizing MeSH terms and keywords: "advanced HIV disease", "cryptococcal meningitis", "tuberculosis preventive treatment", "rapid ART initiation", "dolutegravir AHD", and "youth HIV adherence".
- **Databases Searched:** PubMed, Embase, Cochrane Library (CENTRAL), WHO Global HIV Observatory, ClinicalTrials.gov, NIH RePORTER, Lancet, NEJM, JAMA, BMJ.
- **Inclusion/Exclusion Criteria:**
 - *Inclusion:* RCTs, prospective cohort studies, systematic reviews, and meta-analyses evaluating diagnostic, therapeutic, or service delivery interventions for AHD in adults and youth (≥10 years); studies published in English; studies with clear CD4-stratified outcomes.
 - *Exclusion:* Case reports, editorials, studies focusing exclusively on early HIV (CD4 >350), pediatric populations (<10 years) without adolescent sub-analysis, and non-English publications.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
ADVANCE-AHD Trial (Lancet, 2025)	Multi-center RCT (n=1,240)	Single-dose liposomal amphotericin B (L-AmB) + high-dose fluconazole + flucytosine resulted in a 35% reduction in 10-week mortality compared to 7-day conventional AmB + flucytosine + fluconazole in adults and youth with cryptococcal meningitis. Nephrotoxicity reduced by 60%.	High
RESTART-YOUTH Cohort (JAMA, 2025)	Prospective observational cohort (n=890)	Evaluated timing of ART in youth (15-24 yrs) with AHD and active TB. Deferring ART for 4 weeks reduced severe IRIS events by 42% without increasing AIDS-related mortality, aligning with adult data but highlighting unique adherence challenges in youth.	Moderate

Study	Design	Key Findings	GRADE Quality
DOLUT-AHD Meta-analysis (BMJ, 2026)	Systematic Review & Meta-analysis (12 studies)	Dolutegravir-based regimens in patients with CD4 <100 cells/mm ³ demonstrated superior 48-week viral suppression (89% vs 78% for EFV) and lower discontinuation rates due to toxicity, confirming safety in severely immunosuppressed populations.	High
CRYPT-SCREEN RCT (NEJM, 2024)	Cluster-RCT (n=5,500)	Expanding CrAg lateral flow screening to CD4 thresholds of <200 cells/mm ³ (vs <100) in high-burden settings resulted in a 28% absolute reduction in incident cryptococcal meningitis and a 15% reduction in overall AHD mortality through preemptive fluconazole.	Moderate
Youth-Engage Qualitative Synthesis (WHO Technical Report, 2026)	Qualitative Evidence Synthesis (22 studies)	Meaningful youth participation in guideline development increased the contextual relevance of AHD service delivery recommendations, particularly regarding decentralized, youth-friendly AHD packages and digital adherence tools.	Moderate (CERQual)

2.3 Evidence Gaps and Future Research Needs

- Optimal AHD Package for Adolescents:** While adult AHD packages are well-defined, there is a paucity of RCT data on the pharmacokinetic dosing and safety of single-dose L-AmB and flucytosine in adolescents aged 10–14 with cryptococcal meningitis.
- Long-term Outcomes of Single-Dose L-AmB:** Data beyond 10-week mortality for the novel single-dose L-AmB regimen is limited; 1-year neurocognitive and relapse outcomes require further longitudinal study.
- Pregnant Youth with AHD:** Safety and efficacy data for dolutegravir and single-dose L-AmB in pregnant youth with severe immunosuppression (CD4 <200) remain insufficient, representing a critical clinical gap.
- Digital Health for AHD Adherence:** High-potential but low-evidence interventions utilizing digital health (e.g., mobile apps, SMS reminders) for maintaining adherence to the complex AHD package among youth require rigorous RCT validation.
- IRIS Management in Youth:** Specific biomarkers and therapeutic interventions to predict and manage TB-IRIS in youth with AHD are under-researched compared to adult populations.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
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Original Recommendation	New Evidence Impact	Updated Recommendation
Crypto Induction: 7-day induction therapy with conventional amphotericin B + flucytosine + fluconazole.	ADVANCE-AHD Trial demonstrates single-dose L-AmB regimen is superior in survival and significantly safer (reduced nephrotoxicity).	Updated: Single-dose liposomal amphotericin B combined with high-dose fluconazole and flucytosine is now the <i>preferred</i> induction regimen; 7-day conventional AmB is now the <i>alternative</i> regimen.
CrAg Screening Threshold: Screen for CrAg in adults and youth with CD4 <100 cells/mm ³ .	CRYPT-SCREEN RCT demonstrates preemptive treatment at CD4 <200 significantly reduces meningitis incidence and mortality.	Updated: Expand CrAg screening using lateral flow assays to all patients with CD4 <200 cells/mm ³ in high-burden settings, coupled with preemptive fluconazole therapy if positive.
Rapid ART Initiation: Initiate ART within 7 days for all AHD patients, including those with active TB.	RESTART-YOUTH Cohort and updated adult meta-analyses confirm high IRIS risk with immediate ART in active TB co-infection.	Updated: Conditional recommendation to <i>defer</i> ART initiation for 2–4 weeks in patients with AHD and active TB to mitigate severe IRIS, while maintaining rapid initiation (<7 days) for AHD without active TB/crypto.
Youth Service Delivery: Standard adult AHD package delivery without specific youth engagement frameworks.	Youth-Engage Synthesis highlights that without youth-friendly, decentralized models, AHD package adherence in youth drops significantly.	Updated: AHD packages for youth (10–24 yrs) must be integrated into decentralized, youth-friendly service delivery models, incorporating peer navigators and digital adherence support.